

Practical

FILE PERMISSIONS

Create a file called permission.c

```
lahari@DESKTOP-AIC33F7:~$ nano permission.c
lahari@DESKTOP-AIC33F7:~$ gcc permission.c -o permission
lahari@DESKTOP-AIC33F7:~$ ./permission
File created successfully.
```

Check the current permissions

```
lahari@DESKTOP-AIC33F7:~$ ls -l sample.txt
-rw-r--r-- 1 lahari lahari 28 Aug 17 05:15 sample.txt
```

Give Read and Write permission to everyone

```
lahari@DESKTOP-AIC33F7:~$ chmod 666 sample.txt
lahari@DESKTOP-AIC33F7:~$ ls -l sample.txt
-rw-rw-rw- 1 lahari lahari 28 Aug 17 05:15 sample.txt
```

1) Give Read and Write only to the owner

```
lahari@DESKTOP-AIC33F7:~$ chmod u=rw,go= sample.txt
lahari@DESKTOP-AIC33F7:~$ ls -l sample.txt
-rw----- 1 lahari lahari 28 Aug 17 05:15 sample.txt
```

2) Give Read and Write to the owner and group

```
lahari@DESKTOP-AIC33F7:~$ chmod ug=rw,o= sample.txt
lahari@DESKTOP-AIC33F7:~$ ls -l sample.txt
-rw-rw---- 1 lahari lahari 28 Aug 17 05:15 sample.txt
```

3) Only owner can read and write

```
lahari@DESKTOP-AIC33F7:~$ chmod u=rw,go= sample.txt
lahari@DESKTOP-AIC33F7:~$ ls -l sample.txt
-rw----- 1 lahari lahari 28 Aug 17 05:15 sample.txt
```

4) Owner read/write, everyone else read-only

```
lahari@DESKTOP-AIC33F7:~$ chmod u=rw,o=r,g=r sample.txt
lahari@DESKTOP-AIC33F7:~$ ls -l sample.txt
-rw-r--r-- 1 lahari lahari 28 Aug 17 05:15 sample.txt
```

5) Owner can do everything, others can only read and execute

```
lahari@DESKTOP-AIC33F7:~$ chmod u=rwx,o=rx,g=rx sample.txt
lahari@DESKTOP-AIC33F7:~$ ls -l sample.txt
-rwxr-xr-x 1 lahari lahari 28 Aug 17 05:15 sample.txt
```

6) Owner and group can read/write

```
lahari@DESKTOP-AIC33F7:~$ chmod u=rw,o=,g=rw sample.txt
lahari@DESKTOP-AIC33F7:~$ ls -l sample.txt
-rw-rw---- 1 lahari lahari 28 Aug 17 05:15 sample.txt
```

7) Only owner can execute

```
lahari@DESKTOP-AIC33F7:~$ chmod u=x,o=,g= sample.txt
lahari@DESKTOP-AIC33F7:~$ ls -l sample.txt
---x----- 1 lahari lahari 28 Aug 17 05:15 sample.txt
```

Fill in the permission numbers for the following:

8) Owner: read + write; Group: nothing; Others: nothing → 600

9) Owner: read + write; Group: read; Others: read → 644

10) Owner: read + write + execute; Group: read + execute; Others: read + execute → 755

11) Owner: read + write; Group: read + write; Others: nothing → 660

12) Everyone: read only → 444